



Leaf Disease Detection using Machine Learning

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Problem Statement

In Bangladesh in particular, agriculture plays a vital role in the global economy. Roughly 40% of the working population is employed by it, and it generates 12.5% of the nation's GDP. Over 70% of its land is used to grow major crops such as rice, jute, wheat, tea, lentils, oil seeds, vegetables, and fruits. Plant diseases, however, represent significant hazards to this important economy. Pests and diseases cause up to 40% of the world's agricultural output to be lost each year, which results in an annual economic loss of more than \$220 billion. Plant diseases can have disastrous effects in Bangladesh, wherein a growing middle class has increased demand for premium agricultural products.

Plant diseases may result in smallholder farmers losing a lot of money and impair their ability to produce enough food for themselves or to sell.

Furthermore, in order to feed the world's expanding population by 2050, the Food and Agriculture Organization (FAO) projects that food production will need to rise by around 60% overall and almost 100% in less developed countries. This emphasizes how urgent it is to treat plant diseases that drastically lower food output.

In light of these figures, the goal of our project, "Leaf Disease Detection using Machine Learning," is to provide a reliable and user-friendly method for the early and precise identification of plant diseases. By doing this, we expect to help the global effort to ensure food security and drastically cut the agricultural losses in Bangladesh caused by plant diseases.

WHAT HAVE OTHERS DONE?

Machine learning techniques have been used in a number of studies to detect plant leaf diseases. For example, research that used K-Nearest Neighbor (KNN) classification was able to predict plant leaf diseases with an accuracy of 98.56%. Another research attained great accuracy rates by utilizing deep learning techniques.

Plant leaf disease detection using a variety of machine learning techniques:

- **K-Nearest Neighbor (KNN):** This straightforward yet efficient approach is utilized in several research to identify plant leaf diseases. In one investigation, the prediction of plant leaf diseases was accurate to 98.56%.
- **Support Vector Machines (SVM):** SVMs are very widely utilized in this domain. When there is no linear separability of the data, they are especially helpful.
- **Neural Networks (NN):** These are a class of algorithms that mimic patterns in the human brain. They have been employed to categorize plant diseases in a number of research.
- **Convolutional neural networks, or CNN:** CNNs are a subclass of deep learning neural networks that are mostly used for image analysis. By examining leaf photos, they have demonstrated outstanding effectiveness in the identification of several plant diseases.
- **Deep Learning Techniques:** Compared to more conventional approaches, deep learning has produced significant advances in the field of digital image processing. It has been applied to the research of pest detection and plant diseases.
- **Additional Machine Learning Technologies:** Plant disease classification has also made use of other well-known machine learning algorithms such as Random Forest, Decision Trees, and Naïve Bayes (NB).

WHAT IS THE GAP?

Our study uses innovative methods of machine learning to identify and categorize plant leaf diseases. The project makes use of prototypical networks and few-shot learning (FSL), which work particularly well when there are only a few training examples of a given disease.

But there are some shortcomings of this-

- **Data Quality and Availability:** The quality and availability of data is a significant barrier to the implementation of machine learning models, such as FSL and prototype networks. Despite being extensive, the PlantVillage dataset could not include every kind of leaf disease, particularly those that are unique to a given region.
- **Overfitting:** Given the limited amount of training data and the complicated spectrum characteristic dispersion of the land-cover classes, FSL algorithms are prone to overfit.
- **Computing Requirements:** For training and inference, deep learning models typically require an enormous amount of computing power. This might restrict the application of these models on devices with inadequate processing power.

HOW DO WE OVERCOME THIS?

Overcoming our Shortcomings-

- **Data augmentation:** By expanding both the amount and the variety of the training data, data augmentation techniques may be employed to get around difficulties with data availability and quality.
- **Recurrence Strategies:** Weight decay and dropout are two regularization strategies that can be used to reduce overfitting.
- **Optimized Models:** Less computational resources and parameters are needed when using optimized models, which may assist in satisfying computational requirements.

OUR UNIQUE CONTRIBUTIONS

Unique Contributions of the Project:

- **Use of FSL and Prototypical Networks:** The use of FSL and prototypical networks for leaf disease detection is a novel approach that can handle cases where only a few examples of a particular disease are available.
- **Real-time Disease Detection:** The system could potentially be deployed on an embedded hardware for real-time in-field plant disease detection and identification.
- **Custom Dataset:** The ability to use custom datasets greatly enhances the versatility and applicability of our project. Our model isn't limited to the diseases found in the PlantVillage dataset because it allows for the integration of new datasets. Any user-provided dataset, including ones with pictures of diseases missing from the PlantVillage dataset, can be used to train and test it. Certain plant diseases may be exclusive to a given area. The identification and categorization of these diseases specific to a particular place would be possible with a tailored dataset collected from the surrounding area. With the help of this feature, the model may be trained again using a revised dataset that contains the most recent data, allowing it to learn from new data as it becomes available.
- **High Accuracy:** The proposed implementation has shown high accuracy in predicting plant leaf diseases.